

Practical 13 - Alternative models for survival data

Survival Analysis

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Datasets required: `pbcbase.csv`

R packages required: `survival`, `flexsurv`, `mexhaz`, `timereg`

```
library(survival)
library(flexsurv)
library(mexhaz)
library(timereg)
```

Introduction

In this practical we will return to using the `pbc` data, used in earlier practicals. The data are from a randomised trial and the time origin is the time of randomisation. The variables of interest for this practical are described below.

Variable	Description
<code>id</code>	Unique identifier for each participant
<code>d</code>	Event indicator at the end of follow-up: 0=alive (censored), 1=dead
<code>time</code>	Follow-up time in years
<code>treat</code>	Treatment: 1=placebo, 2=active
<code>bil0</code>	Serum bilirubin (mg/dl), measured at the start of the trial.

The practical is in 3 parts:

- Part A: Flexible parametric models
- Part B: Accelerated failure time models

- Part C: Additive hazards models

Load the data using:

```
pbcb<-read.table("datasets/pbcbase.csv",sep=" ",header=T)
```

Aims

The aims of this session are:

- Fit flexible parametric models (FPM) using a spline for the baseline hazard, and obtain corresponding survival curves.
- Fit Weibull models and make the link between proportional hazards and accelerated failure time (AFT) parametrizations.
- Fit Weibull and log-logistic AFT models and compare the fit.
- Fit additive hazard models and interpret the graphical output.

Part A: Flexible parametric models

Question 1

In this question we will fit a flexible parametric model (FPM) to the PBC data, with the baseline hazard modelled using a spline (as a function of time).

Use the code below to carry out the following steps: - Start by calculating the 25th, 50th and 75th percentiles (i.e. the quartiles) of the log follow-up time for individuals who have the event. We call these `myknots`. - Using `flexsurvspline` fit a FPM including `treat` and `bil0` as covariates. For the baseline hazard use a natural cubic spline with 3 interior knots for $\log(t)$ located at the quartiles of log follow-up time in people who have the event. This can be done by specifying `knots=myknots` and `timescale="log"` (this is the default for `timescale`). - The same thing can be done using `flexsurvspline` and simply specifying the number of knots as `k=3`, which will place 3 interior knots at the quartiles of $\log(t)$ in people who have the event. - Take a look at what the output is showing and link this to what was shown in the slides.

```
#obtain quartiles of log(t)
myknots<-quantile(log(pbc$time[pbc$d==1]),probs=c(0.25,0.5,0.75))

#fit FPM using above knot placements
pbc.flex1<-flexsurvspline(Surv(time,d)~treat+bil0,data=pbc,
                          knots=myknots,timescale="log",spline="rp")
```

```

pbc.flex1

#same thing using k=3
pbc.flex2<-flexsurvspline(Surv(time,d)~treat+bil0,data=pbc,
                          k=3,timescale="log")
pbc.flex2

```

Question 2

- Use the code below to obtain estimated survival curves based on the FPM fitted in question 1 for two individuals: person 1 has `treat=1`, `bil0=30` and person 2 has `treat=2`, `bil0=30`.
- Fit a new FPM with 10 internal knots instead of 3. Compare the coefficients from the two models.
- Compare estimated survival curves from the two models for the two individuals above.

```

#Obtain estimated survival; probabilities at times 0,0.1,0.2,...,11.9,12
surv.flex1<-predict(pbc.flex1,
                   newdata=data.frame(treat=c(1,2),bil0=c(30,30)),
                   type="survival",times=seq(0,12,0.1))

#Plot survival curves
times<-seq(0,12,0.1)
surv.person1.flex1<-surv.flex1$.pred[[1]]$.pred_survival
surv.person2.flex1<-surv.flex1$.pred[[2]]$.pred_survival

plot(times, surv.person1.flex1, type="l",lwd=2,
     xlab="Time",ylab="Survival probability")
lines(times, surv.person2.flex1, type="l",lwd=2,col="grey")
legend(x="topright",c("Treat=1,bil0=30","Treat=2,bil0=30"),
     lty=1,lwd=2,col=c("black","grey"),cex=0.8)

```

Question 3

- Compare your hazard ratio estimates from the FPM (with 3 knots) with those from using a Cox proportional hazards model.
- Compare the estimated survival curves for the two individuals considered above.
- What are the assumptions of the FPM versus the Cox model?

Part B: Accelerated failure time models

Question 1

- Use `mexhaz` to fit a Weibull model with treatment group as the explanatory variable, using the proportional hazards parametrization.
- Write down the model being fitted and interpret your results.

```
#Weibull model using PH model
mod.weib.ph=mexhaz(Surv(time,d)~as.factor(treat),data=pbcc,
                   base="weibull")
summary(mod.weib.ph)
```

Question 2

- Use `survreg` to fit a Weibull model with treatment group as the explanatory variable, using the AFT parametrization. Note that `survreg` uses the AFT parametrization for the Weibull model by default.
- Write down the form of the survival function under the AFT Weibull model and interpret your results.
- What is the estimate of the acceleration factor $e^{\beta_{AFT}}$? Interpret the estimate.
- What is the relationship between your estimates of the hazard ratio $e^{\beta_{PH}}$ and the acceleration factor $e^{\beta_{AFT}}$?

```
#Weibull model using AFT model
mod.weib.aft=survreg(Surv(time,d)~as.factor(treat),data=pbcc,
                    dist="weibull")
summary(mod.weib.aft)
```

Question 3

- A log-logistic model is another type of AFT model. Use `survreg` to fit a log-logistic model with treatment as the explanatory variable.
- Find the estimates of the acceleration factor, and interpret the results.
- Compare the fit of the Weibull model and the log-logistic model using the AIC. Which model provides the better fit?

```
#Log-logistic model
mod.loglog=survreg(Surv(time,d)~as.factor(treat),data=pbcc,
                    dist="loglogistic")
summary(mod.loglog)

#AICs from the Weibull and log-logistic models
AIC(mod.weib.aft)
AIC(mod.loglog)
```

Part C: Additive hazard models

Question 1

- Fit the additive hazards model using `treat` and `bil0` as explanatory variables.
- Write down the form of the model fitted and take a look at the estimated coefficients.
- Plot the cumulative coefficients and interpret the graphical output.

```
#Fit additive hazards model
mod.aalen=aareg(Surv(time,d)~as.factor(treat)+bil0,data=pbcc)

#The coefficients are in here:
#The first column (actually the rownames here) is the times
head(mod.aalen$coefficient)

#Plot the cumulative coefficients
par(mfrow=c(1,3))
plot(mod.aalen)
```

Question 2

In this question we will compare survival curves estimates using the additive hazards model and using the Cox model.

The code below takes you through the following steps:

- We first need to use a different function for fitting the additive hazards model (`aalen` in the `timereg` package) because using `aareg` in the `survival` package does not allow us to automatically obtain survival estimates.

- Based on the additive hazards model fitted above, obtain the estimated survival curves for individuals 1 and 2: person 1 has `treat=1`, `bil0=30` and person 2 has `treat=2`, `bil0=30`.
- Compare the estimated survival curves with those obtained from the Cox model in Part A.

```
#Fitting the model using timereg package.
#The resample.iid=1 option is needed if we want to later use `predict`.
mod.aalen=aalen(Surv(time,d)~as.factor(treat)+bil0,data=pbcc,
                 resample.iid = 1)

#The cumulative coefficients are in here:
head(mod.aalen$cum)

#Plot the cumulative coefficients
par(mfrow=c(1,3))
plot(mod.aalen)

#Obtain survival estimates
surv.aalen<-predict(mod.aalen,
                    newdata=data.frame(treat=c(1,2),bil0=c(30,30)))
surv.person1.aalen<-surv.aalen$S0[1,]
surv.person2.aalen<-surv.aalen$S0[2,]

#Plot survival curves, and compare with those from the Cox model
par(mfrow=c(1,1))
plot(surv.aalen$time, surv.person1.aalen, type="s",lwd=2,
     xlab="Time",ylab="Survival probability")
lines(surv.aalen$time, surv.person2.aalen, type="s",lwd=2,col="grey")
lines(surv.cox$time, surv.person1.cox, type="s",lwd=2,col="black",lty=2)
lines(surv.cox$time, surv.person2.cox, type="s",lwd=2,col="grey",lty=2)
legend(x="topright",
      c("Aalen: Treat=1,bil0=30","Aalen: Treat=2,bil0=30",
        "Cox: Treat=1,bil0=30","Cox: Treat=2,bil0=30"),
      lty=c(1,1,2,2),lwd=2,
      col=c("black","grey","black","grey"),cex=0.8)
```